

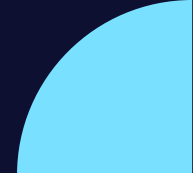


Gates, Circuits & Algorithms

From qubits to real computation

[Person 2] · Big Data seminar
Quantum Computing · Part 2 of 3





Roadmap



01

Gates

Reversible moves and a small toolkit

02

Circuits

Entanglement, and the real trick

03

Grover

Search in about $\sqrt{N}$ steps

04

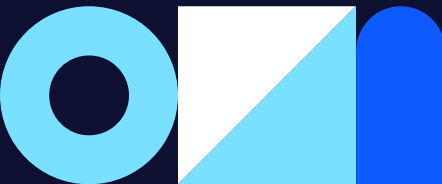
Shor

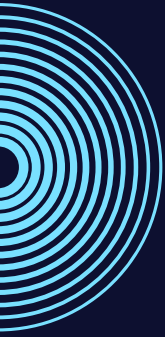
Factoring and the crypto clock

05

Live demo

Grover's search, run twice

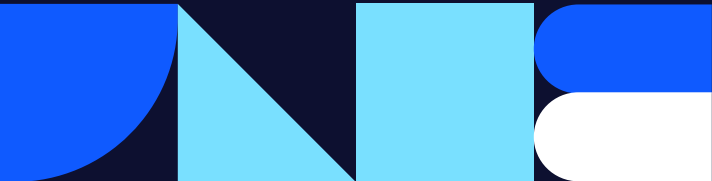




01

Quantum gates

How do qubits change without losing information?



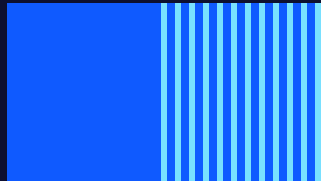


Rule 1: gates can be undone

A classical AND gate loses information: an output of 0 could come from 00, 01 or 10, so you can't go back.

A quantum gate never does: n qubits in, n qubits out, and you can always run it backwards. The one step you can't undo is **measurement**.

In math terms, gates are “unitary” matrices (more on slide 9).



Flip the bit, flip the sign

X gate

Quantum NOT: swaps $|0\rangle$ and $|1\rangle$.

Y gate

Does both at once: flips the bit and the sign.

Z gate

Sign flip: $|1\rangle$ gets a minus sign, $|0\rangle$ is untouched.

The minus sign is invisible at measurement, but it decides what cancels later.

Hadamard: the coin flip

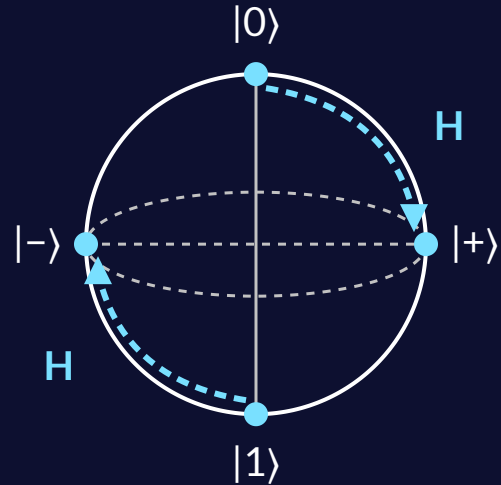
50 / 50

H turns a certain $|0\rangle$ into an equal superposition

H, then H

brings $|0\rangle$ back for certain (a real coin flipped twice stays random!)

Almost every algorithm starts with H on every qubit: all inputs at once, equal weight.



H moves $|0\rangle$ from the north pole to the equator: a 50/50 superposition.



CNOT links two qubits

If the first qubit is 1, flip the second.

Input	Output	What happens
00	00	First is 0: nothing happens
01	01	First is 0: nothing happens
10	11	First is 1: flip the second
11	10	First is 1: flip the second

Give it a 50/50 first qubit and the pair is **linked**: $(|00\rangle + |11\rangle)/\sqrt{2}$. Measure one, and you know the other.





A tiny toolkit builds it all

Classical

NAND alone builds any
circuit

Quantum

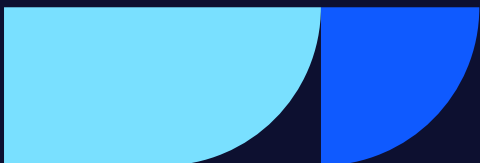
H, CNOT and two phase
gates build any program

Toffoli

Reversible AND: quantum
can run any classical
program

The catch

Can build anything is not
the same as fast



The math behind gates

- Gates are unitary: $U^\dagger U = I$, so probability is preserved and $U^{-1} = U^\dagger$
- CNOT: $|c, t\rangle \rightarrow |c, t \oplus c\rangle$
- $\{H, S, T, \text{CNOT}\}$ approximates any operation to any accuracy ($S = T^2$)
- Solovay-Kitaev: single-qubit gates need only $O(\log^c(1/\epsilon))$ gates, $c \approx 2$; arbitrary n -qubit operations can still need exponentially many

Hadamard	$\text{---} \boxed{H} \text{---}$	$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$
Pauli-X	$\text{---} \boxed{X} \text{---}$	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
Pauli-Y	$\text{---} \boxed{Y} \text{---}$	$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$
Pauli-Z	$\text{---} \boxed{Z} \text{---}$	$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
Phase	$\text{---} \boxed{S} \text{---}$	$\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$
$\pi/8$	$\text{---} \boxed{T} \text{---}$	$\begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$

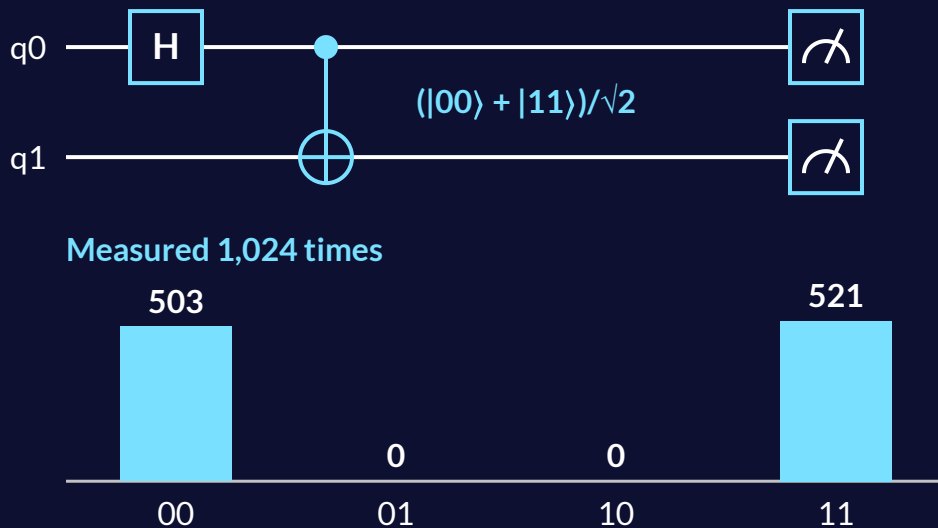
Source: Nielsen & Chuang (2010), Fig. 4.2.



02 Circuits

Chain gates together and watch entanglement appear

Two gates make entanglement



503 · 521

shots on 00 and 11; 01 and 10 never appear

H makes the first qubit 50/50,
CNOT links the second to it:
 $(|00\rangle + |11\rangle)/\sqrt{2}$.

One line per qubit, time runs left to right; the dot is the CNOT control, $\oplus$ the target.



All inputs at once?



3 qubits $\rightarrow$ 8 inputs

53 qubits $\rightarrow \approx 9 \times 10^{15}$ inputs

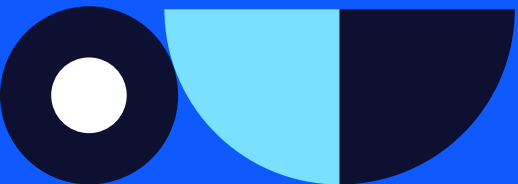




The catch

**Measuring gives back one
random answer. Parallelism
alone is not the trick.**

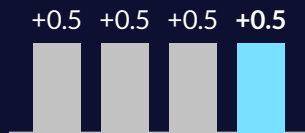
So what is the trick? Interference.





Make wrong answers cancel

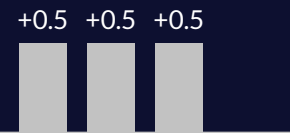
1 • Start



00 01 10 11

every answer +0.5

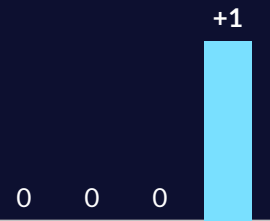
2 • Mark 11



00 01 10 11

11 becomes -0.5

3 • Amplify



00 01 10 11

11 → +1, rest → 0

100%

on 11 after one amplify step

Amplitudes can be negative, so they add up or cancel, like waves.
Not “try everything at once”:
engineered cancellation.

2 qubits, looking for 11. The amplitude of each answer at each stage.





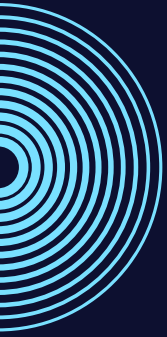
One check, not 524,289

A hidden function is either “always the same” or “half 0, half 1”. Which one?

Method	Checks	Answer
Classical	524,289	Up to this many, to be certain (20-bit inputs)
Quantum	1	Interference gives “all zeros” (always the same) or “never all zeros” (half and half)

A teaching example, not a practical speedup: random classical checks are nearly always right.

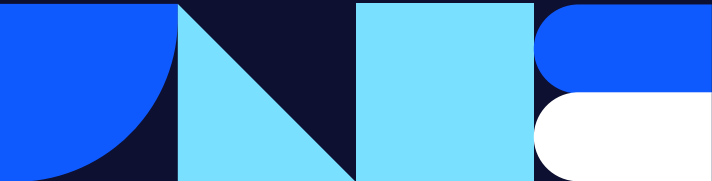




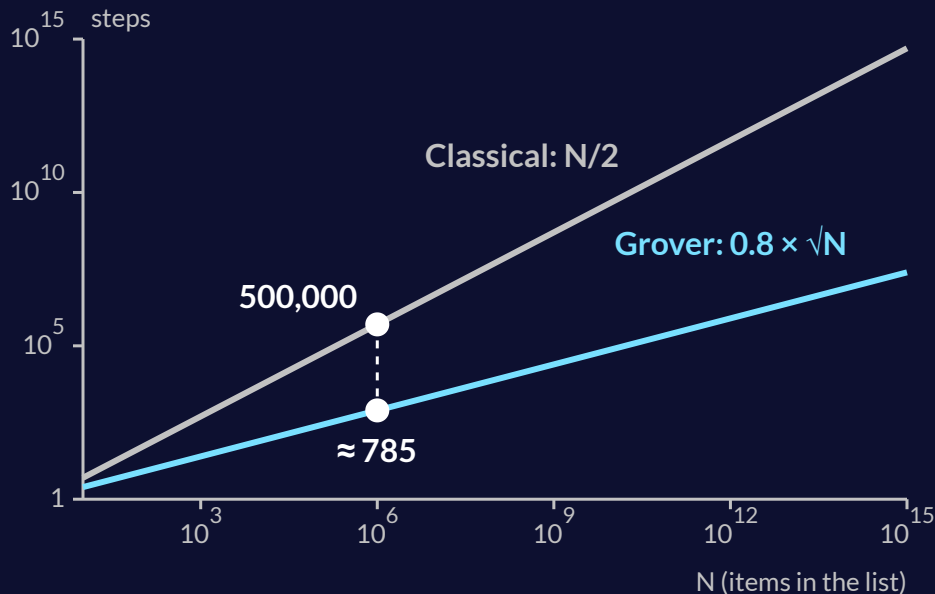
03

Grover's search

N items, no order, no index, one is special. Find it.



Grover: $\sqrt{N}$ instead of N



Steps needed as the list grows (log-log). Classical $N/2$, Grover about $0.8 \times \sqrt{N}$.

500,000

checks for 1,000,000 items, classical

≈ 785

Grover steps for the same list

Proven best possible for this problem. A big speedup, but not exponential.

Each round: mark, then amplify

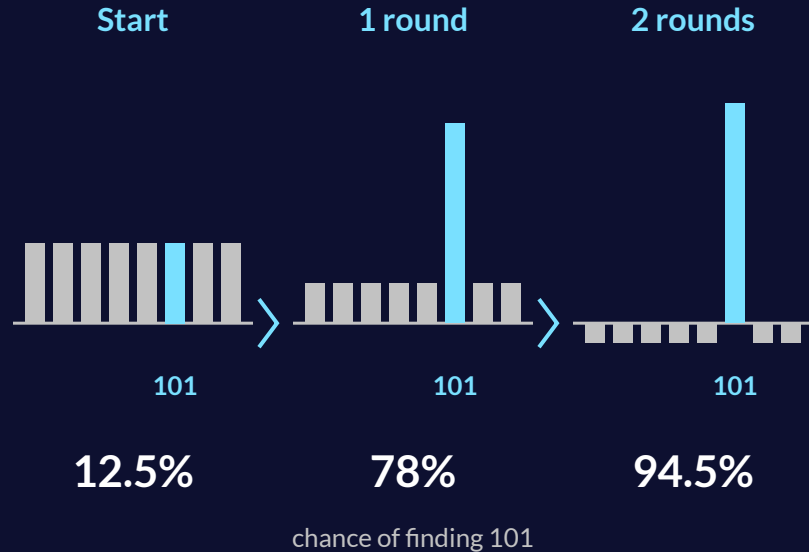
Mark (oracle)

A checker puts a minus sign on the answer, without saying which one.



Amplify (diffusion)

Flip every amplitude around the average: the marked one jumps up, the rest shrink.



Rotate, and stop in time

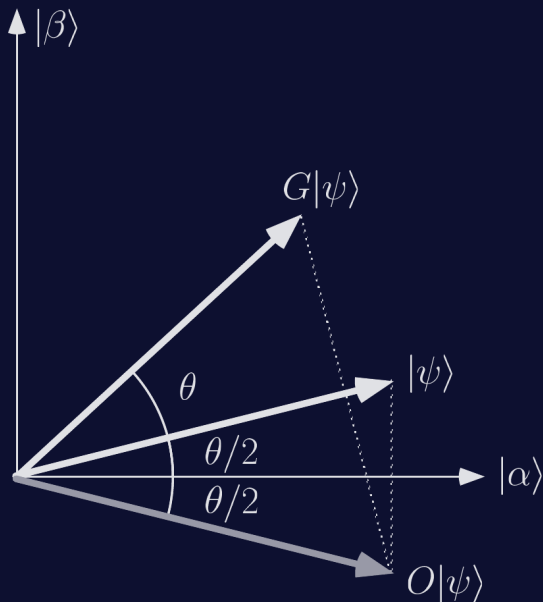
$$0.8 \times \sqrt{N}$$

rounds is about the best place to stop

$$94.5\% \rightarrow 33\% \rightarrow 1\%$$

for 8 items after 2, 3 and 4 rounds: too many rounds overshoot

You don't need the answer to know when to stop, only how many items there are.



Source: Nielsen & Chuang (2010), Fig. 6.3. The book starts at $\theta/2$ and rotates by θ ; we say θ and 2θ .



How many rounds?

Best number of rounds is about $(\pi/4)\sqrt{N}$. Success after k rounds is $\sin^2((2k+1)\theta)$.

Items N	Best rounds	Success
4	1	100%
8	2	94.5%
16	3	96.1%
100	7	99.5%

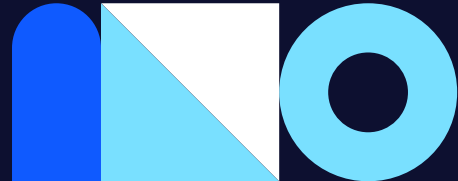




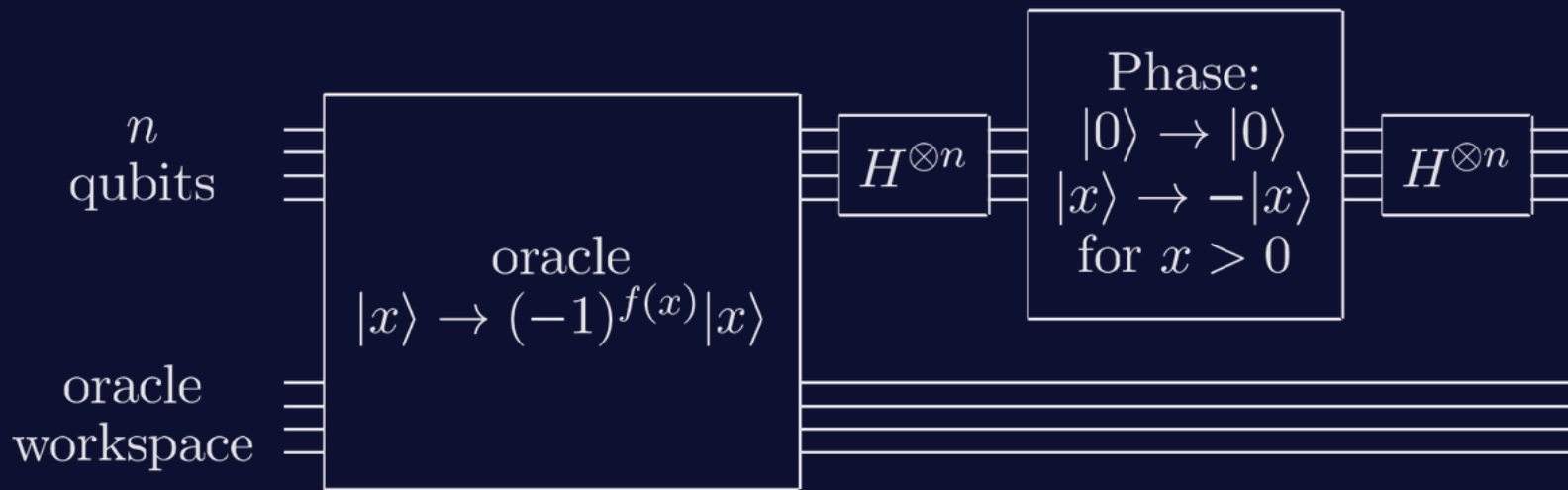
The Grover formulas

The numbers behind the search:

- Start angle θ with $\sin \theta = 1/\sqrt{N}$
- Each round turns the state by 2θ
- Diffusion reflects about the mean: $a \rightarrow 2 \cdot \text{mean} - a$
- 3 qubits, amplitude of 101: $0.354 \rightarrow 0.884 \rightarrow 0.972$
- No quantum algorithm beats order $\sqrt{N}$ (Bennett et al., 1997)



One Grover round



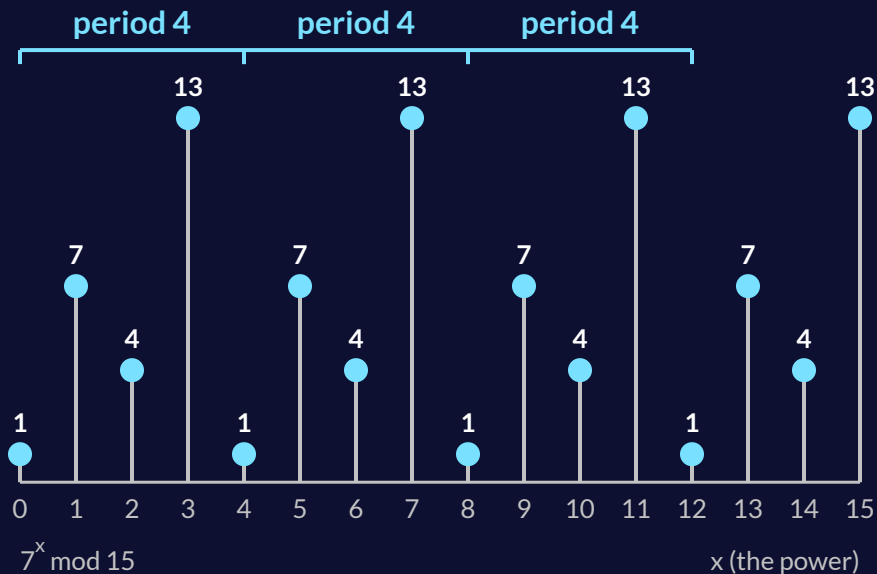
Source: Nielsen & Chuang (2010), Fig. 6.2. The demo's H-X-CZ-X-H diffusion adds a global minus sign, which never changes measurements.



04 Shor & crypto

Why factoring matters for security

Factoring = finding a rhythm



Powers of 7, remainder after dividing by 15. The pattern repeats every 4 steps.

7, 4, 13, 1

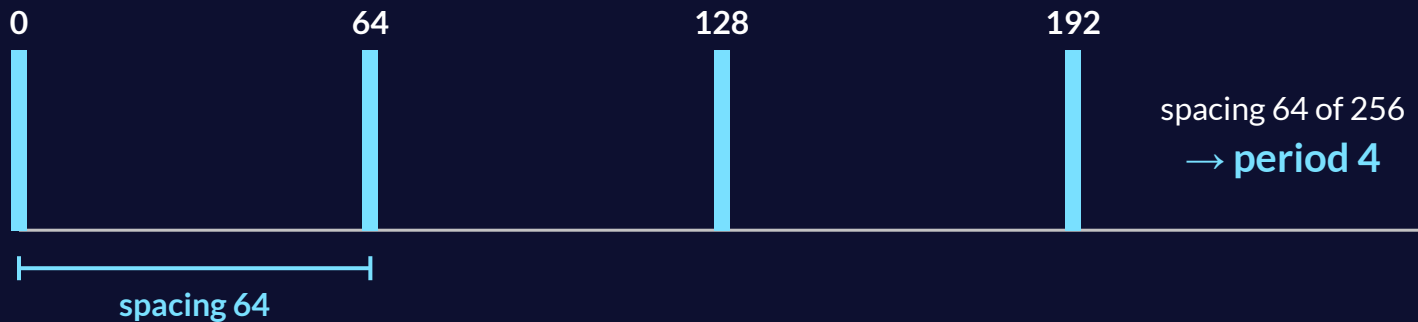
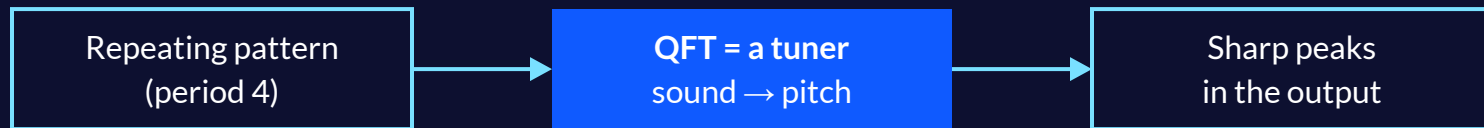
remainders of powers of 7 divided by 15, then it repeats

15 = 3 × 5

the period 4 hands you the factors

For a 2,048-bit number the pattern is astronomically long.

The QFT: a tuner for rhythms



Classical: cost explodes with digits · Shor: only polynomial



Shor by the numbers

01

Period

$f(x) = a^x \bmod N$
repeats with period r .

02

Conditions

r even and $a^{r/2} \not\equiv -1 \pmod{N}$: true for at least half of random a .

03

Factors

$\gcd(a^{r/2} \pm 1, N)$. For 15: $\gcd(48, 15) = 3$, $\gcd(50, 15) = 5$.

04

QFT cost

$n(n+1)/2$ gates on n qubits. Its output sits in amplitudes, not directly readable.





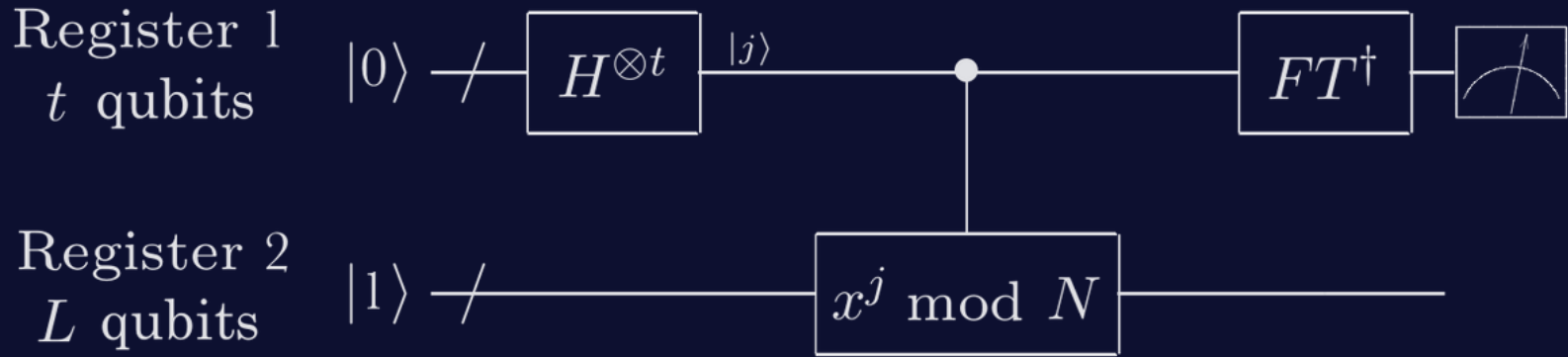
The cost of factoring

n = number of bits in N .

Method	Cost
Best classical (number field sieve)	$\exp(O(n^{1/3} (\log n)^{2/3}))$
Shor	about $O(n^2 \log n \log \log n)$



The order-finding circuit



Source: Nielsen & Chuang (2010), Fig. 5.4. Counting qubits, controlled modular powers, inverse QFT, then measurement.



What Shor breaks

Shor attacks the public-key part of HTTPS and digital signatures.

Breaks	Survives	The fix
RSA Elliptic-curve crypto Diffie-Hellman	AES Hash functions Just use longer keys (AES-256)	NIST post-quantum standards (2024): ML-KEM, ML-DSA, SLH-DSA





The crypto clock

RSA-2048: about 20 million
noisy qubits

2019

Under 1 million qubits,
under a week

2025

Start migrating long-lived
data

Next

— **01** — **02** — **03** — **04** — **05** —

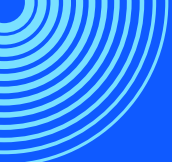
2024

NIST post-quantum
standards published

Today

No machine is close yet





The big-data angle

**Every TLS link, encrypted data
lake and signed pipeline is on
the list.**

Harvest now, decrypt later: migrate long-lived data now.





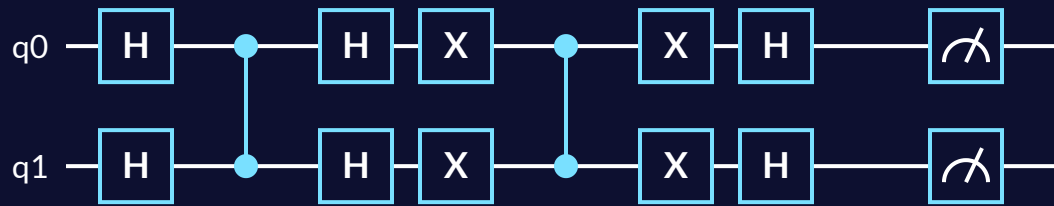
05

Live demo

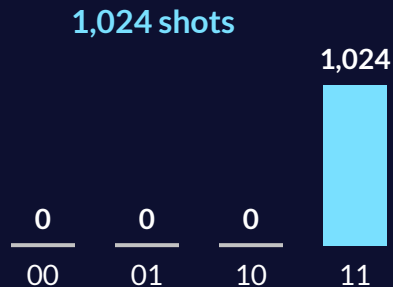
Grover's search, run twice



Demo 1: find 11 among 4



H on both → mark 11 → amplify → measure

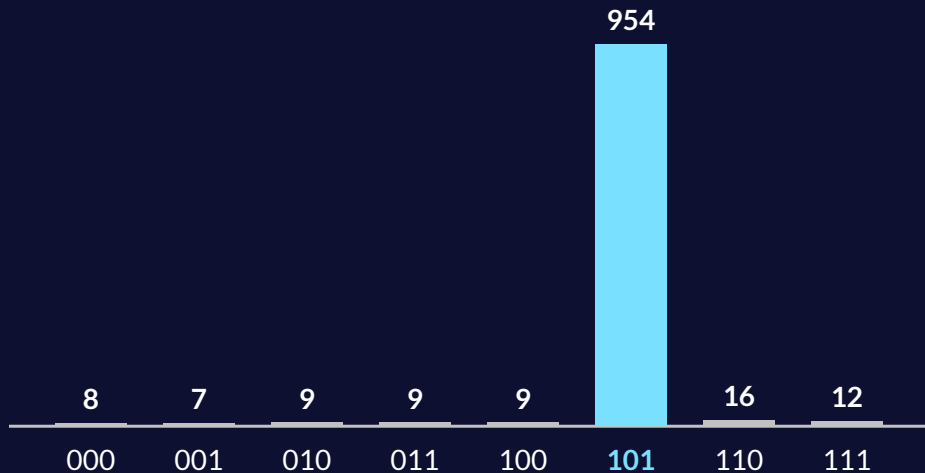


Theory: 1 round, 100% on 11

Result: {'11': 1024}

every shot correct

Demo 2: find 101 among 8



93.2%

= 954 of 1,024 shots on 101

94.5%

theory, after 2 rounds

The other seven states get about 1% each. Random shots wobble by about ± 0.7 points.

3 qubits, looking for 101. Counts from 1,024 shots (seed 7), the same run Part 3 shows.

What we built

Gates

Reversible moves; a small toolkit is enough

Circuits

Two gates make entanglement

Interference

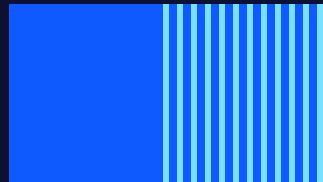
The real trick: wrong answers cancel

Grover

Search in $\sqrt{N}$ steps, the best possible

Shor

Easy factoring, so we need post-quantum crypto





Thank you!

Next: what does this buy a big-data system, and what hardware
exists today?

Part 3 · Quantum × Big Data

